Customer Shopping Behaviour Analysis Using SQL

Portfolio Findings Report

Dataset: Customer Shopping Trends Dataset

Tools: SQLite and DB Browser for SQLite

Scope: 3,900 shopping records

This report summarises an independent SQL analysis of recorded purchase amounts and shopping behaviour. It is a practice portfolio project using a public dataset, not work completed for a real client.

Project objective

Explore purchase amounts across product categories and customer groups, compare payment methods and reported shopping frequency, perform selected data quality checks, and communicate findings with appropriate limitations.

1. Executive Summary

Measure	Finding
Records analysed	3,900
Total recorded purchase value	$233,081
Average purchase amount	$59.76
Minimum / maximum purchase amount	$20 / $100
Observed age range	18–70

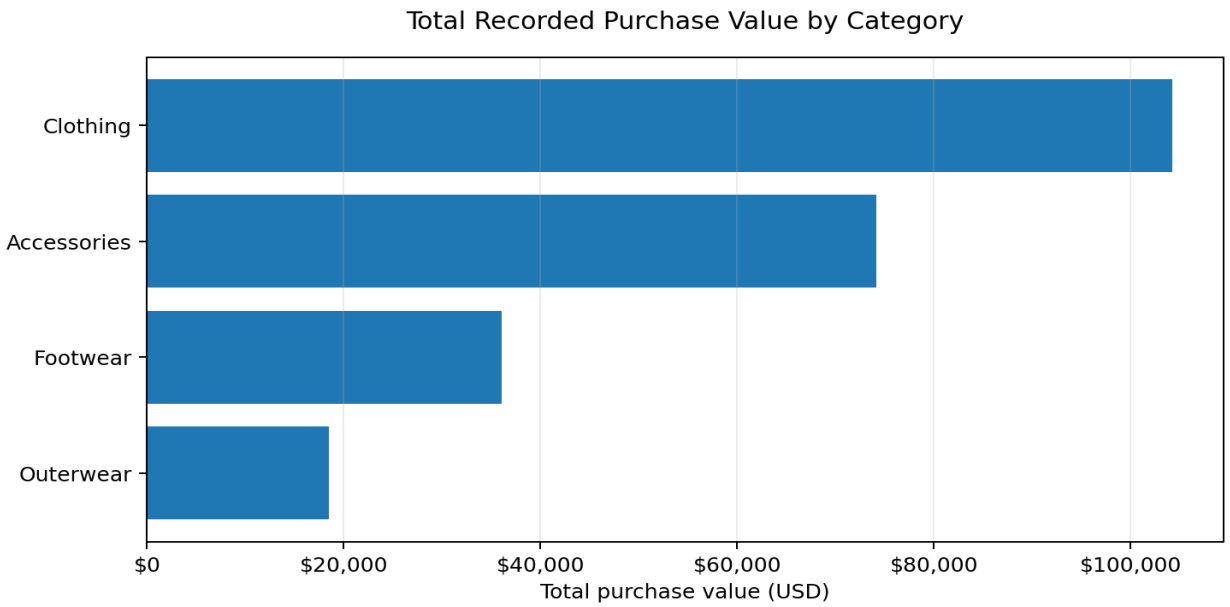
• Clothing accounts for the largest number of records and the highest total recorded purchase value.

• Footwear has the highest average purchase amount by category, but its total value is much lower than Clothing's.

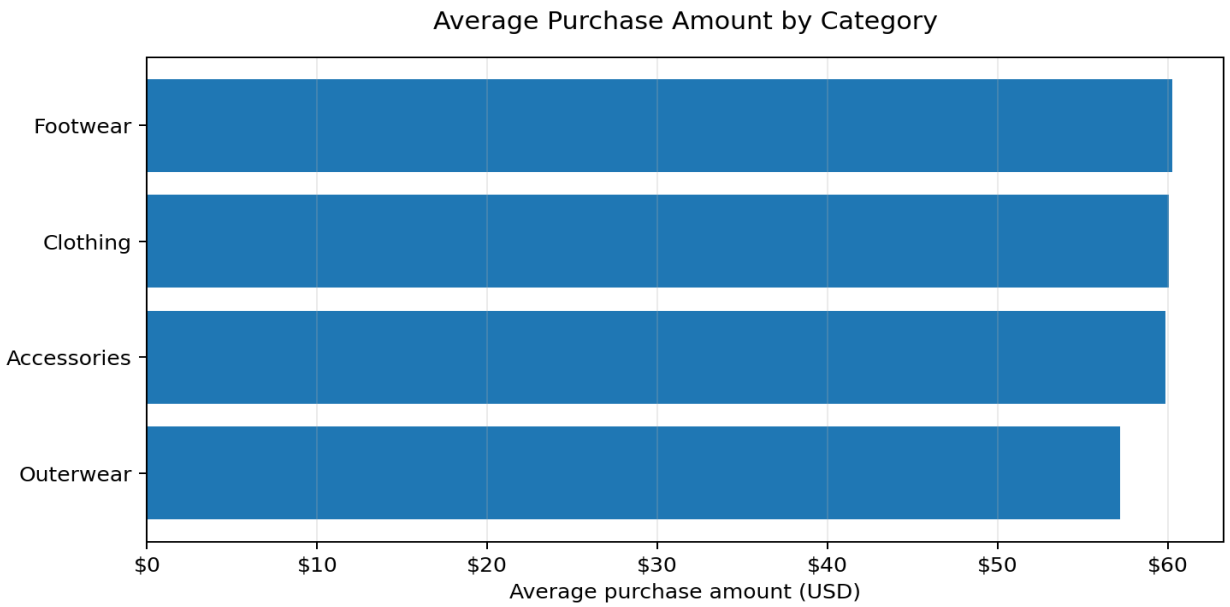
• Group differences should be interpreted alongside record counts; averages alone do not establish which group is most valuable.

• The dataset does not include costs or repeated dated transactions, so it cannot establish profitability or customer lifetime value.

2. Product Category Findings



Clothing recorded $104,264 in total purchase value across 1,737 records, the highest among the four categories. Accessories followed with $74,200 across 1,240 records. Footwear recorded $36,093 and Outerwear $18,524.



Footwear had the highest average purchase amount (about $60.26), while Outerwear had the lowest (about $57.17). A higher average does not necessarily mean a higher total value or greater profit.

3. Customer Group Findings

Age group	Records	Average amount	Total value
Young Adult	878	$60.26	$52,905
Adult	1,463	$59.31	$86,766
Older Adult	1,559	$59.92	$93,410

Young Adults had the highest average purchase amount ($60.26), while Older Adults had the highest total recorded purchase value ($93,410) and the most records. These are different measures and should not be treated as interchangeable.

Gender comparison

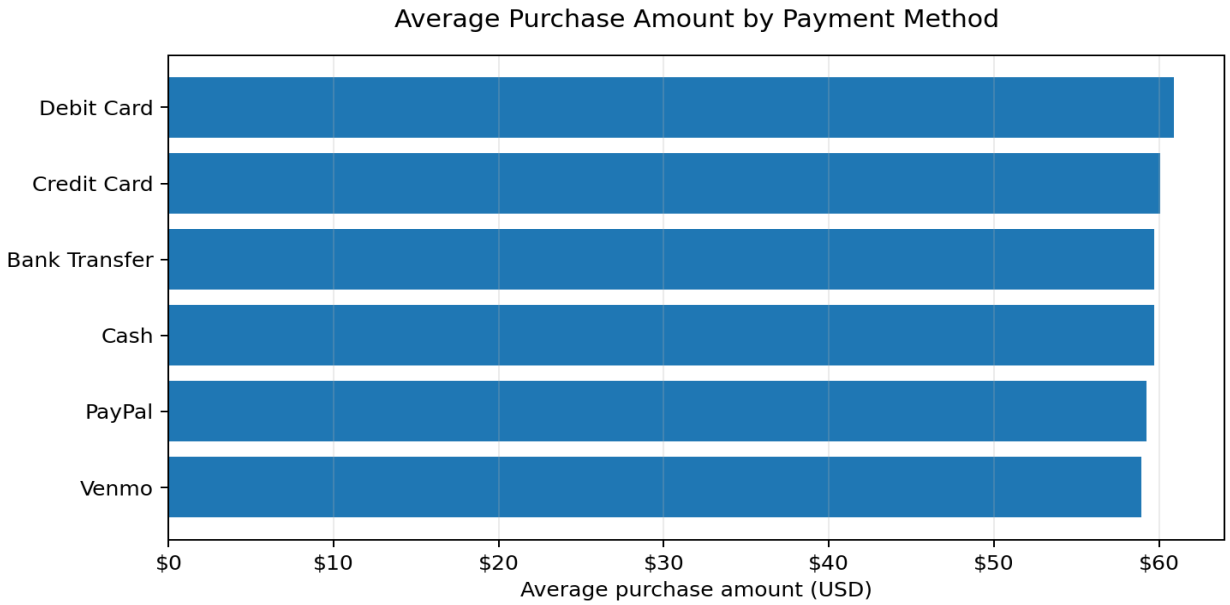
Female records averaged about $60.25, compared with $59.54 for male records. Male records had a much higher total value ($157,890 versus $75,191), alongside a larger number of records (2,652 versus 1,248). This comparison does not explain why the groups differ.

4. Subscriptions and Discounts

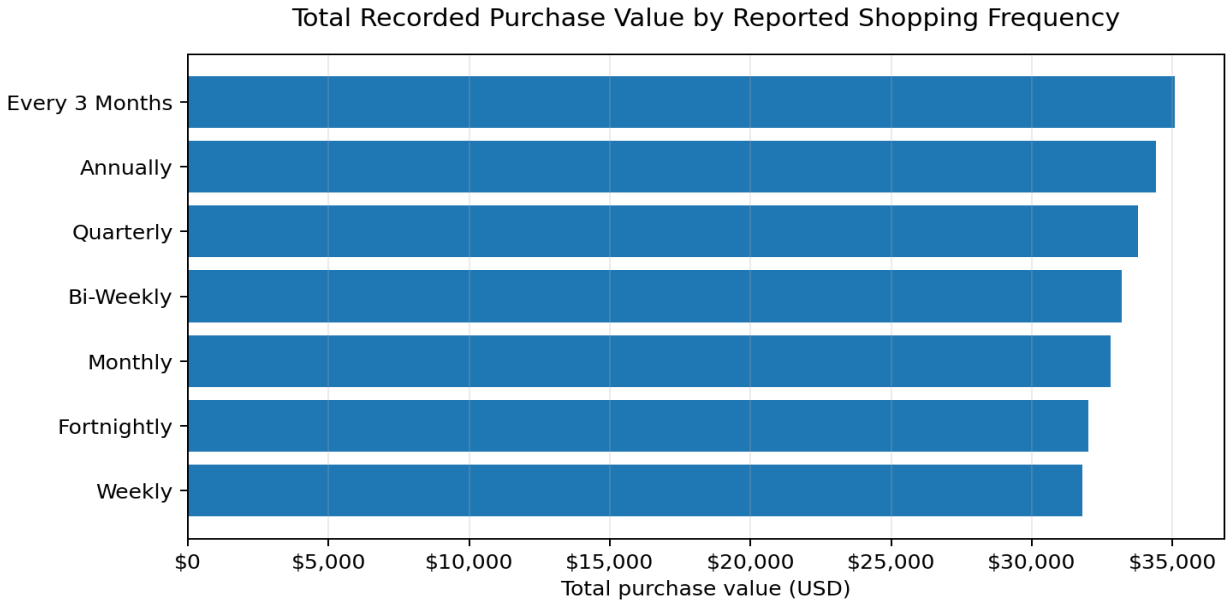
Comparison	Records	Average amount	Total value
No subscription	2,847	$59.87	$170,436
Subscription: Yes	1,053	$59.49	$62,645
No discount	2,223	$60.13	$133,670
Discount applied	1,677	$59.28	$99,411

Records without discounts had a higher average purchase amount than records with discounts. This does not prove discounts reduce spending: customer mix, product mix, and other factors may differ between groups. Subscription groups also had similar average review ratings (about 3.75 for No and 3.74 for Yes).

5. Payment Methods and Shopping Frequency



Across all records, Debit Card had the highest average purchase amount ($60.92), while Venmo had the lowest ($58.95). Credit Card had the highest total recorded purchase value ($40,310). A payment method's average and total value can rank differently.



The “Every 3 Months” group had the highest total recorded purchase value ($35,088), and the “Weekly” group had the lowest ($31,786). Record counts across frequency groups were relatively close. Because each Customer ID appears only once, reported frequency is not a verified history of repeated purchases.

6. Data Quality Checks

• 3,900 records and 3,900 distinct Customer IDs; no duplicate IDs were found.

• No missing values were found in the selected checks for purchase amount, review rating, and subscription status.

• No purchase amounts at or below zero were found.

• The observed age range was 18–70; no records had age below 18.

• No review ratings outside the 1–5 range were found.

7. Limitations and Responsible Interpretation

• Each Customer ID appears once, so the dataset has one record per customer rather than repeated dated transaction history.

• Recorded purchase value is not profit. Cost, margin, operating expenses, and customer acquisition costs are not available.

• Group comparisons are descriptive and do not establish cause and effect.

• Totals are affected by group size; review averages, record counts, and total values together.

• “Quarterly” and “Every 3 Months” may be similar labels, as may “Bi-Weekly” and “Fortnightly.” The analysis kept the labels separate.

• The findings describe this dataset and should not automatically be generalised to every retailer.

8. Conclusion

The analysis demonstrates practical SQL skills in data exploration, data quality checks, aggregation, filtering, group comparisons, conditional grouping with CASE, and subqueries. The strongest business lesson is to compare averages, totals, and group sizes together and to state limitations clearly before making recommendations.

Dataset source: Customer Shopping Trends Dataset, Kaggle — https://www.kaggle.com/datasets/iamsouravbanerjee/customer-shopping-trends-dataset